

Assignment 2

Draftosaurus

This project will allow you to develop a computer game with simple graphics and with the support of AI. The game is a drafting game in which players build a zoo of dinosaurs. The aim of the project is not simply to produce a working version of the game; it is also to take you through the key stages of software development: design, coding, testing, and debugging.

This project is worth 7.5% of your final grade. There are three tasks, each with allocated marks. You must attempt all tasks on your own before seeking help from AI. The final submission deadline is the last teaching day of semester, Friday 16th October 2026 at 5pm, although some of the marks will be gained in the last three programming tutorials. Your submission will be marked in the last three tutorials, in the lab, and also your C# code and a final report will be marked after you submit your program to Moodle.

Project Specification:

Design and implement a computer version of the “Draftosaurus” game by Team Kaedama. The game involves collecting dinosaurs for your zoo and organising them into pens in a way that maximises your score.

Your program should implement a two-player version of the game, where the user plays either against the computer or against another player. You can find the rules and a description of the two-player version at <https://boardgamegeek.com/boardgame/264055/draftosaurus>

The game must satisfy the specifications described above and follow the provided game rules. Your program should make use of the programming features you have learnt so far, including classes and methods to structure your code, as well as XML documentation where appropriate.

We may ask you to explain your code while marking your program in the lab. Therefore, you should review your code carefully and make sure that you understand how it works, particularly any code generated with the help of AI.

You must also document all AI prompts used during the project, as marks will be awarded for this.

Tasks

Week 10: Due in your programming tutorial during week 10 (Oct 1 or Oct 2)

Read through the game rules and write a summary of the game. Consider the problem in terms of its constraints, behaviours, and the information that needs to be stored. Analyse your summary to identify the game objects (nouns), their behaviours (verbs), and how they interact with one another (relationships). Later, these game objects may become suitable object-oriented programming classes, the identified behaviours may become methods, their interaction becomes object-oriented programming relationships e.g. inheritance, composition and aggregation.

Design the classes required for your game and draw a UML class diagram. Present your diagram in your programming tutorial for discussion. Your class diagram will be marked during the tutorial.

Week 11: Due in your programming tutorial during week 11 (Oct 8 or Oct 9)

Get AI to design the classes and data structures. Compare this with your class design. Note what is different and think about what that difference means. Decide whether your class diagram, or the AI class diagram, needs to be updated, or both.

Think about how you are going to interact with your game, designing your user interface and describing the user actions and interactions.

Present at your programming tutorial a list of tests you will need to make to see if your program is working correctly. Specifically, what tests should happen about choosing a dinosaur to play, and once a dinosaur has been chosen, what tests should happen about how you (can) play that dinosaur? Bring this list and your user interface design to your programming tutorial to discuss. Your lists and user interface design will be marked during the tutorial.

Week 12: Due in your lab during week 12 (Oct 12 to Oct 16)

Use AI to build your game based on your class diagram and other design notes. Record the prompts you use to generate the code, as well as any further prompts or refinements needed to improve the AI-generated code.

Test your AI-built game using your prepared list of test cases. Record the test results and any additional tests that you find necessary. Summarise how closely your AI-generated game follows the rules provided in the rulebook.

In your lab in week 12, show your working program to a demonstrator for verification.

Final Handin: Due 5pm Friday 16th October

- A project report containing three sections.

Section One should include your original class diagram, the AI-generated class diagram, and a discussion of the differences between them. You should also present your final class diagram and justify the decisions you made in developing it.

Section Two should include your interface design, your original testing scheme, and the AI-improved testing scheme. Discuss and reflect on the differences between the two testing schemes and justify any modifications you made.

Section Three should provide a summary of how you used AI and evaluate its effectiveness in designing the class diagram, generating code, and testing and debugging the program. You should also discuss how closely your AI-assisted game follows the rules provided in the rulebook.

You must include all prompts used with AI as evidence.

- A zip or 7z of your C# game code.

Hand this report (.docx or pdf), along with a zip or 7z of your C# game code, into Moodle by 5pm on the last day of term.

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Assignment 2 Hand-in due 5pm 16th October 2026

Compress (Zip or 7z) the Visual Studio folder with your program code and submit it via Moodle, along with your final report in either .pdf Word or .docx format.

Student Declaration of Originality

I declare that the program which I have had verified and submitted in Moodle is entirely my own work, alongside the AI used. I have not worked with any other people. I have suitably acknowledged (referenced) all AI generated parts. I understand that if I have breached the above conditions, I will be sent to the University Disciplinary Committee.

Name: _____ **ID:** _____

Signed: _____ **Date:** _____

Class Diagram (shown in the week 10 tutorial) _____ /4 marks

	Poor	Average	Good	Excellent
Object design, inheritance, composition	0	1	2	2
UML Diagram	0	1	2	2

Testing Scheme (shown in the week 11 tutorial) _____ /4 marks

	Poor	Average	Good	Excellent
Testing Scheme: list of test conditions, outcomes	0	1	2	2
Usability: layout, choice of controls, feedback	0	1	2	2

Functionality and Usability (demonstrated in the lab) _____ /4 marks

	Minimal or none	Partial, or glitchy	Works
Game setup, randomness and hand swapping	0	½	1
Player can place a dino in pen or discard	0	½	1
Placement restrictions enforced (area, size, T-Rex)	0	½	1
Game end & scoring (pens, river, T-Rex) correct	0	½	1

Report (marked after zip and report submitted) _____ /8 marks

	Poor	Average	Good, Excellent
Reflection of object design changes after AI	0	1	2
Reflection of testing, with results and updates	0	1	2
Documentation of AI used and prompts	0	1	2
Summary of game correctness and AI effectiveness	0	1	2

Total: _____ /20 marks